

Refining ChatGPT’s Analysis on $\omega_{\dagger 1}$

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Abstract

We aim to analyze OpenAI’s ChatGPT¹ 5.0’s ability to reason with novel mathematical concepts. Specifically, an idea developed by Aytzhanov of well-orderings subject to “sort irreducibility”. With LLMs becoming increasingly proficient in existing mathematics, one naturally wonders if LLMs are capable enough to deal with niche mathematical ideas such as $\omega_{\dagger 1}$. GPT will be instructed to analyze whether $\omega_{\dagger 1}$ exists and its potential ramifications, including inconsistency. Then, its answers will be continuously refined via carefully constructed response prompts. This gives insight into the future of LLMs in mathematics/computer science research as we force ChatGPT to reason with ideas with very little literature surrounding them, and these ideas’ applications expand far into various fields.

1 Background

We will ask ChatGPT about a niche mathematical concept in the field of “mathematical foundations”. Such a field requires airtight reasoning and thus would typically be difficult for a LLM to reason through. Additionally, the fact that this is a niche topic means GPT cannot search the internet for answers. In general, foundations has many applications in mathematics and CS, so if ChatGPT were to be successful in this endeavor, then it would indicate ChatGPT’s possible effectiveness in independent research.

1.1 Mathematical Information

The crux of the problem that we will query GPT about will now be developed.

¹Any and all LLM response text will be properly labeled with ORANGE. This style was created by Johannes Schmitt.

Definition 1.1. A logical sort is a “kind of object”, or a unique domain to quantify over.

More set theories, such as ZFC are “1-sorted” theories. They only operate upon sets of a single nature, and thus are “1-sorted”.

Definition 1.2 (Aytzhanov). Denote the least well ordering not amenable to any β -sorted structure $\omega_{\dagger\beta}$.

Lemma 1.3. *The supremum of all well orderings amenable to a singular logical sort is $\omega_{\dagger 1}$.*

Our question is, does $\omega_{\dagger 1}$ even exist? Before asking about the higher properties of $\omega_{\dagger 1}$ or how it behaves in various models, we must ask ourselves if such a structure is even logically possible. Of course, we must decide our framework. Regular ZFC does not permit us to even speak of such structures effectively, but we should at least force our theory to be some sort of extension of ZFC (preferably conservative).

Before we can analyze the response of the AI, we must ask ourselves what the correct answer to this question is.

Theorem 1.4. *$\omega_{\dagger 1}$ does not exist.*

Proof. Obviously, $\omega_{\dagger 1}$ is not amenable to a singular sort, so it must be at least 2-sorted. Suppose it exists and is n -sorted for the sake of contradiction. We can describe any n -sorted structure $(V_1, V_2, V_3, \dots, V_n, \in)$ with a 1-sorted structure

$$\left(\bigcup_{i < n} \{i\} \times V_i, \in^* \right).$$

This will allow us to create an isomorphism between the two structures as long as we choose $\in^*$ correctly, which is easy. Thus, we can collapse any n -sorted structure down to 1-sort, meaning that no matter where $\omega_{\dagger 1}$ “lives”, it can be reduced back down to 1-sort. Thus $\omega_{\dagger 1} < \omega_{\dagger 1}$, contradiction. $\square$

This argument, in essence is a response to the alleged solution $\omega_{\dagger 1}$ gives to the Burali-Forti paradox. There are two solutions to this any given Burali-Forti paradox, and for this one they are:

1. $\omega_{\dagger 1}$ does not exist.

2. $\omega_{\uparrow 1}$ is not 1-sorted.

Our above proof shows that the second solution is not possible, and thus $\omega_{\uparrow 1}$ does not exist.

2 Analysis

Our initial prompt will define any relevant ideas and ask our main question. We will give the following initial prompt to GPT:

HUMAN | A theory has n logical sorts if there are n “kinds of objects” in that theory. For example, ZFC has one sort because it only has sets and that is its only domain. We will work in some extension of ZFC (possibly multi-sorted or with proper classes). Let $\omega_{\uparrow 1}$ be the supremum of all well orderings amenable to a singular logical sort. $\omega_{\uparrow 1}$ thus could not be expressed in a singular logical sort. Does $\omega_{\uparrow 1}$ exist? Limit your response to about a paragraph.

ChatGPT gives the following response:

AI | In any reasonable foundational setting (including ZFC, NBG, MK, or multi-sorted extensions thereof), $\omega_{\uparrow 1}$ does not exist as an ordinal or set, and not even as a well-defined proper class in a robust sense. The reason is essentially a Burali–Forti–type phenomenon combined with a meta-theoretic defect: the notion “amenable to a singular logical sort” is not absolute or internal to the theory, but depends on how the theory is presented and coded. Any attempt to take the supremum of all well-orderings expressible in a single sort either (i) yields a proper class that cannot itself be well-ordered within that same expressive regime, or (ii) collapses into contradiction, since the supremum would itself define a new well-ordering, thereby exceeding the purported bound. Thus $\omega_{\uparrow 1}$ cannot exist as a determinate object; at best it is an informal meta-concept indexing expressive strength rather than a genuine ordinal.

GPT correctly identifies the core idea, but does not correctly elaborate on any of the details of the proof. Furthermore, it gives us several ideas on the perspective of $\omega_{\uparrow 1}$ instead of just 1. We will further probe it to elaborate in a chain-of-thought-esque style where we ask it to expand on its most important ideas.

HUMAN | Give a detailed mathematical argument, step by step, to support your most important idea on $\omega_{\uparrow 1}$ ’s (non)existence.

AI I will make precise the core obstruction, which is a **Burali–Forti–style diagonal argument relativized to expressive power**.

1. Fix a theory T with a single logical sort S (one domain of quantification), possibly with parameters. Inside T , define $\mathbf{WO}_1$ to be the class of all well-orderings on subsets of S that are definable in T (or otherwise “amenable” in your sense) using only that one sort.
2. **Each such well-ordering has an order type**, which—if it exists at all—can be represented internally as an ordinal (or externally as a class ordinal). Let

$$\mathcal{O}_1 = \{\alpha : \alpha \text{ is the order type of some } W \in \mathbf{WO}_1\}.$$

3. Suppose, towards contradiction, that $\omega_{\dagger 1} = \sup \mathcal{O}_1$ exists as a well-defined ordinal or class ordinal.
4. By definition of supremum, $\omega_{\dagger 1}$ is itself a well-ordering of its initial segments. But crucially, the construction of $\omega_{\dagger 1}$ depends only on quantifying over $\omega_{\dagger 1}$, which itself is definable using a single logical sort (since “being a definable well-ordering on S ” is a one-sorted notion).
5. Therefore, the ordering $(\omega_{\dagger 1}, \in)$ is *amenable to the same singular sort*. Hence its order type must lie in $\mathcal{O}_1$.
6. But then $\omega_{\dagger 1} \in \mathcal{O}_1$, contradicting the fact that $\omega_{\dagger 1}$ strictly dominates every element of $\mathcal{O}_1$.
7. This contradiction shows that **no such supremum can exist**. The failure is structural, not technical: any attempt to “close off” all one-sorted well-orderings immediately generates a new one of strictly greater type, definable at the same expressive level.

Conclusion. $\omega_{\dagger 1}$ cannot exist as an ordinal or class ordinal. It is an inherently unstable meta-concept: the expressive act of defining it already exceeds the boundary it is meant to capture. This mirrors Burali–Forti, but the diagonalization is over *expressibility*, not merely over ordinals.

GPT now correctly applies a Burali–Forti style argument, but we are not done yet. While

the argument might seem robust, we must also consider that it rests upon some (reasonable) assumptions. However, GPT may not be able to correctly identify that these assumptions to be reasonable, if queried. Thus, we will continue to “ask questions” about the argument.

HUMAN

I have a few questions. First, what if we permit T to be multi-sorted and force $\omega_{\dagger 1}$ to also be multi-sorted? Is there also a contradiction here? Second, why doesn't this type of argument rule out regular infinite ordinals?

AI

(i) **Allowing T to be multi-sorted does not rescue $\omega_{\dagger 1}$**

The contradiction does not come from insisting that $\omega_{\dagger 1}$ live in the same syntactic sort as the well-orderings it bounds, but from requiring that it collect all well-orderings amenable to a fixed expressive regime. Suppose T has many sorts and includes a distinguished sort S representing “one-sorted objects.” Let WO_S be the class of well-orderings definable using only S . Even if $\omega_{\dagger 1}$ is placed in a separate sort S' , its definition must still quantify over WO_S . That quantification yields a canonical well-ordering of order type $\sup \mathcal{O}_S$, and the relation defining that ordering is again expressible using only S -quantification. Hence one obtains a new well-ordering amenable to S that strictly exceeds every element of $\mathcal{O}_S$, reproducing the same diagonal contradiction. The extra sort merely hides the construction; it does not block it.

(ii) **Why this does not rule out regular infinite ordinals.**

The key distinction is that regular ordinals (e.g. ω , ω_1) are internal closure points of already-given collections (such as unions of smaller ordinals), not “suprema of all objects definable at a given expressive level.” Their defining property—regularity—does not involve quantifying over *all ordinals definable in the theory*, but only over subsets or sequences indexed by smaller ordinals. No expressive self-reference is involved. By contrast, $\omega_{\dagger 1}$ is defined by ranging over all well-orderings available to a given logical sort, which is a meta-theoretic totality unstable under its own definition. Thus regular infinite ordinals are safe, while $\omega_{\dagger 1}$ fails for the same reason Burali–Forti fails: it tries to totalize a notion that regenerates itself at the same expressive level.

From here, we attempt to steer GPT into the original proof argument given above. We do this by forcing it to think about abandoning the nonexistence solution and to steer it toward that second “solution”. Our hope is that GPT does not fall for this trap.

What if we insist $\omega_{\dagger 1}$ must be an object that operates over at least 2-sorts? For example, for theory T with sorts S and S' , $\omega_{\dagger 1}$ would be an order-type forcibly containing elements from both S and S' (and it would be defined as the smallest such order-type that has to have elements from both).

Requiring $\omega_{\dagger 1}$ to *live across* at least two sorts changes the surface syntax but not the underlying obstruction. One can make this precise as follows.

Fix a theory T with two sorts S and S' . Let WO_S be the class of all well-orderings definable using only quantification over S . Now define a *two-sorted* well-ordering to be a relation $R(x, y)$ on $S \cup S'$ such that (i) R well-orders the disjoint union $S \sqcup S'$, and (ii) both sorts occur nontrivially. Suppose $\omega_{\dagger 1}$ is defined to be the least order-type of such a two-sorted well-ordering which strictly dominates every order-type in WO_S .

The key point is that this still yields a diagonalization. From $\omega_{\dagger 1}$, restrict R to the S -part of its domain. Because R is a well-order of the whole union, its restriction to S is a well-order of S . Moreover, the definition of this restriction uses only the S -sort: membership in S is primitive, and no quantification over S' is needed. Hence the induced order-type lies in WO_S .

But since the full two-sorted order strictly dominates every element of WO_S , its S -initial segment must have order-type at least $\omega_{\dagger 1}$, contradicting the fact that $\omega_{\dagger 1}$ was supposed to be strictly larger than all one-sorted order-types. Put differently: forcing the global order to “mention” a second sort does not prevent its S -part from being a new one-sorted well-ordering of unbounded type.

Conceptually, the problem is invariant under adding sorts: as long as the definition of $\omega_{\dagger 1}$ is meant to *dominate all well-orderings accessible to a fixed expressive fragment*, any object that witnesses this domination canonically generates—by restriction, coding, or projection—a new well-ordering still accessible to that fragment. Thus even a forcibly multi-sorted $\omega_{\dagger 1}$ cannot exist as a stable order-type; the expressive boundary it is meant to cap is not closed under its own supremum.

Indeed, GPT did not fall into the trap. It even correctly identifies to use the disjoint union $\sqcup$, which is equivalent to what we did.

So, can you collect all of this into a one paragraph argument about why your claim about $\omega_{\dagger 1}$ holds for a theory (ideally some extension of ZFC) with n sorts? In other words, create a general argument/proof.

AI Fix a theory T extending ZFC with finitely many sorts $S_1, \dots, S_n$, and fix one distinguished sort S_1 . Let $\mathbf{WO}_{S_1}$ denote the totality of all well-orderings whose domains and defining relations are expressible using only quantification over S_1 . Suppose, for contradiction, that there exists an object $\omega_{\dagger 1}$ (possibly living across several sorts) which is the supremum of the order-types of all members of $\mathbf{WO}_{S_1}$, in the sense that it strictly dominates each of them. Because the definition of $\omega_{\dagger 1}$ necessarily quantifies over $\mathbf{WO}_{S_1}$, it canonically induces a well-ordering of type $\omega_{\dagger 1}$; restricting this ordering to the S_1 -part of its domain yields a well-ordering definable using only S_1 , hence an element of $\mathbf{WO}_{S_1}$. But this restricted order must have order-type at least $\omega_{\dagger 1}$, contradicting the assumption that $\omega_{\dagger 1}$ strictly exceeds every order-type in $\mathbf{WO}_{S_1}$. This diagonal contradiction is invariant under adding finitely many sorts or forcing $\omega_{\dagger 1}$ to be multi-sorted: any object that purports to bound all S_1 -expressible well-orderings inevitably regenerates a new S_1 -expressible well-ordering of equal or greater type. Hence no such $\omega_{\dagger 1}$ can exist as a set, class, or stable order-type in any finite-sorted extension of ZFC.

At this point, we have reached a satisfactory answer. Comparing the initial answer to the end, the difference is clear. ChatGPT was able to successfully develop a convincing argument with regards to the nonexistence of $\omega_{\dagger 1}$ ², a mathematical object with very little literature attached to it. The best strategy concerning research with AI is to probe it by asking about corner cases as well as asking it to explain itself clearly.

3 Conclusion

This endeavor was insightful into a LLM's potential future applications into research. ChatGPT has shown its capabilities in enumerating through various ideas and applying them. A researcher must still prompt and guide the exploration, but ChatGPT is capable in thinking of a lot of the findings itself. One particular challenge seen was getting the AI to adhere to the constraints of the problem properly. GPT often went off course or did not assume the proper preconditions. One particular idea of improvement is to create custom research-oriented GPT variants that follow through specifically with formal proofs and that value the user's instructions more.

²It is also worth mentioning that ChatGPT also agrees with the proof we developed, although this part of the conversation (which was done strictly *after* the above) is not shown.